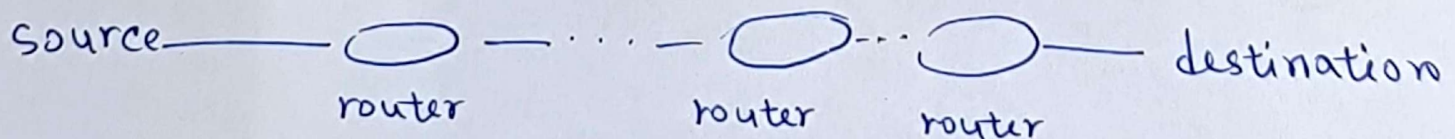
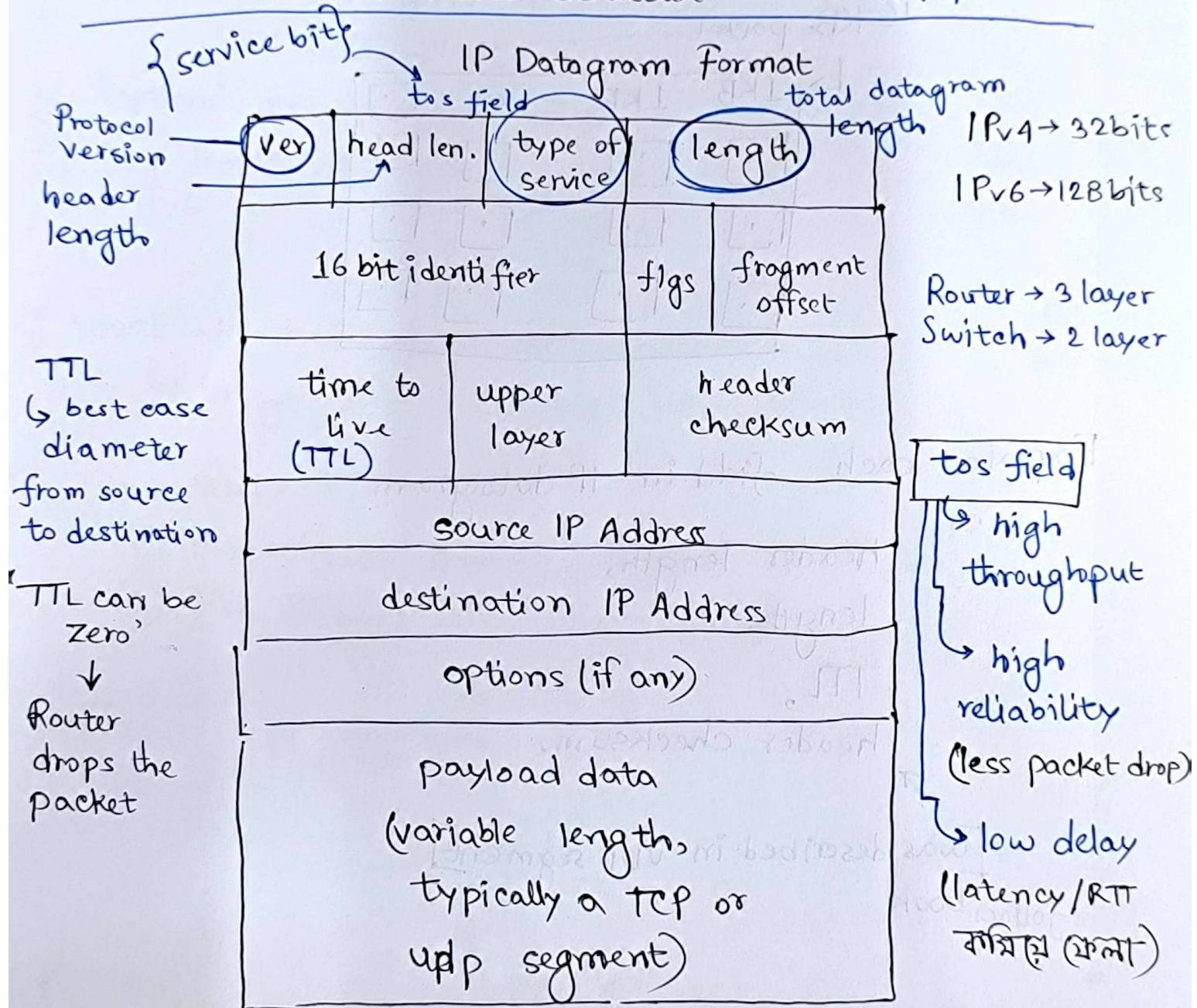


#Chapter Four

IPv4 and IPv6 Packet Structure

IPv4



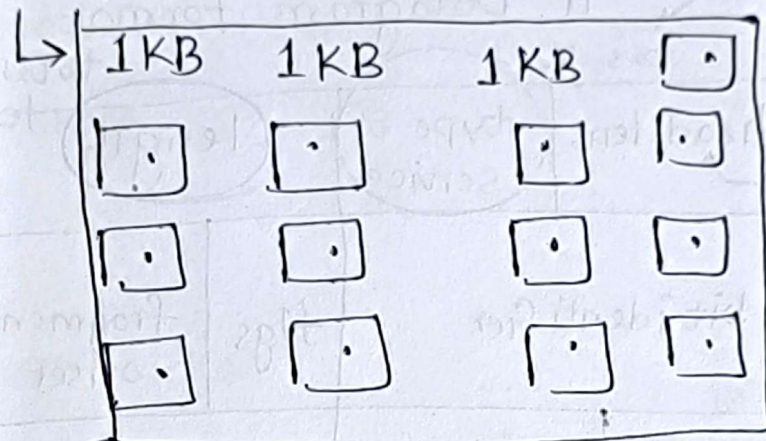
$$\text{max length} \rightarrow (2^{16} - 1) \approx 64 \text{ KB}$$

✓ TCP থাকবে upper layer '6'

✓ UDP থাকবে " " '17'

fragment → packet ভেঙে ছোট ছোট করা

16 KB packet



Explain each field in IP datagram

or

header length,

length,

TTL,

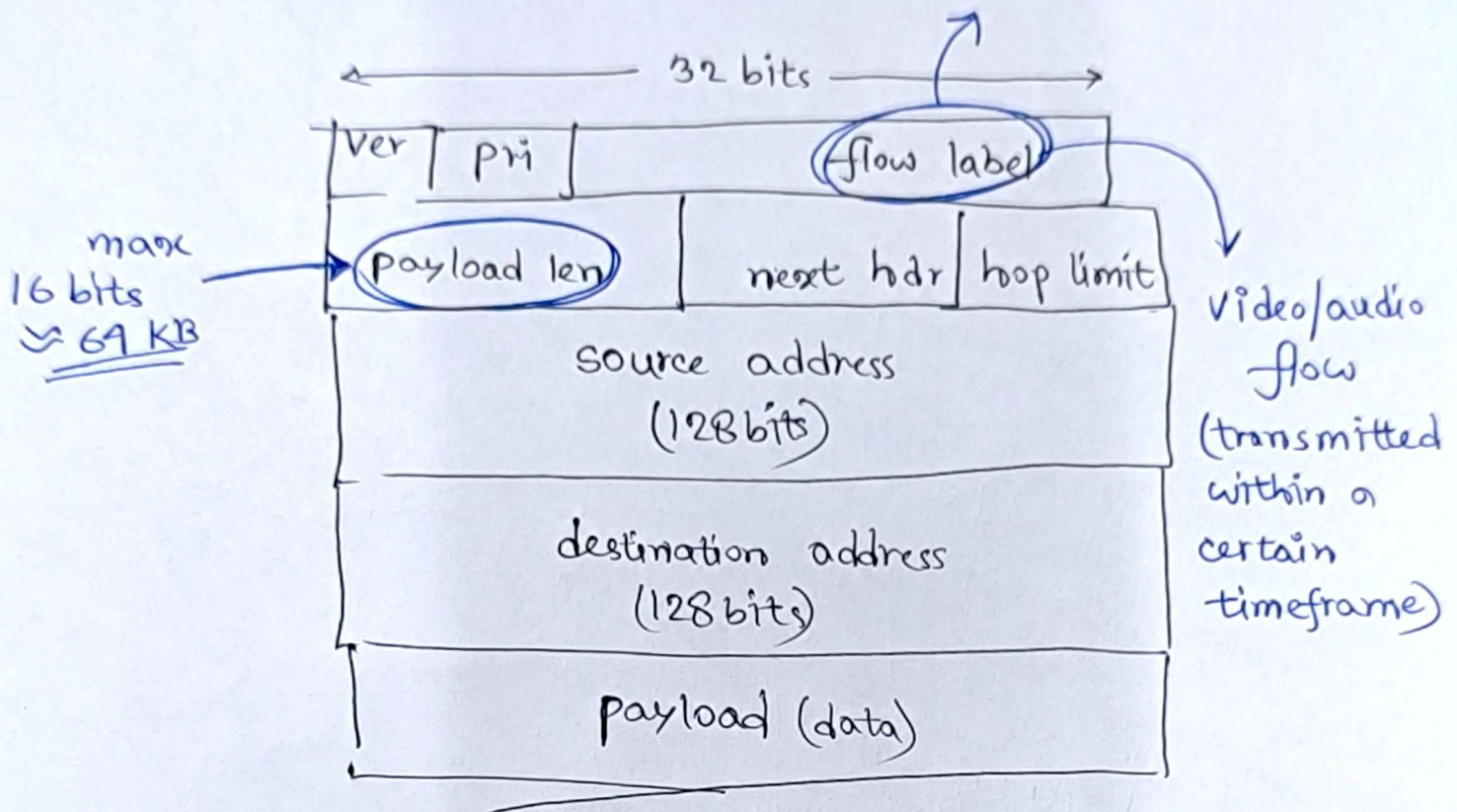
header checksum.

[was described in UDP segment]

follow book

IPv6

can be categorized



❌ 'Options' are dropped in IPv6

✅ 'hop limit' equivalent of 'TTL'
IPv6 IPv4

✅ 'next hdr' → "Next Header"

↳ Integer value → changes with protocol

✅ flow label:

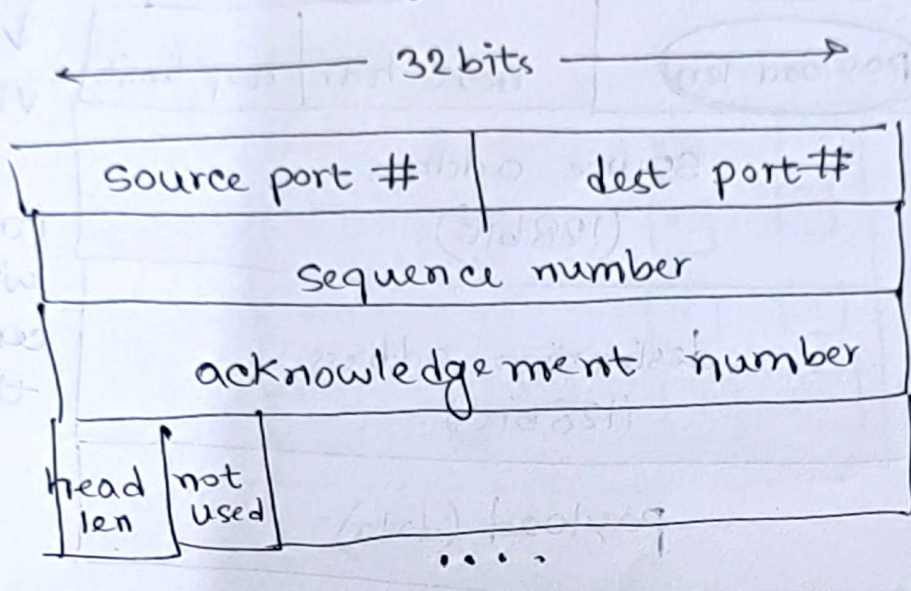
① Video/Audio transfer <time constraint>

② File Transfer <no time constraint>

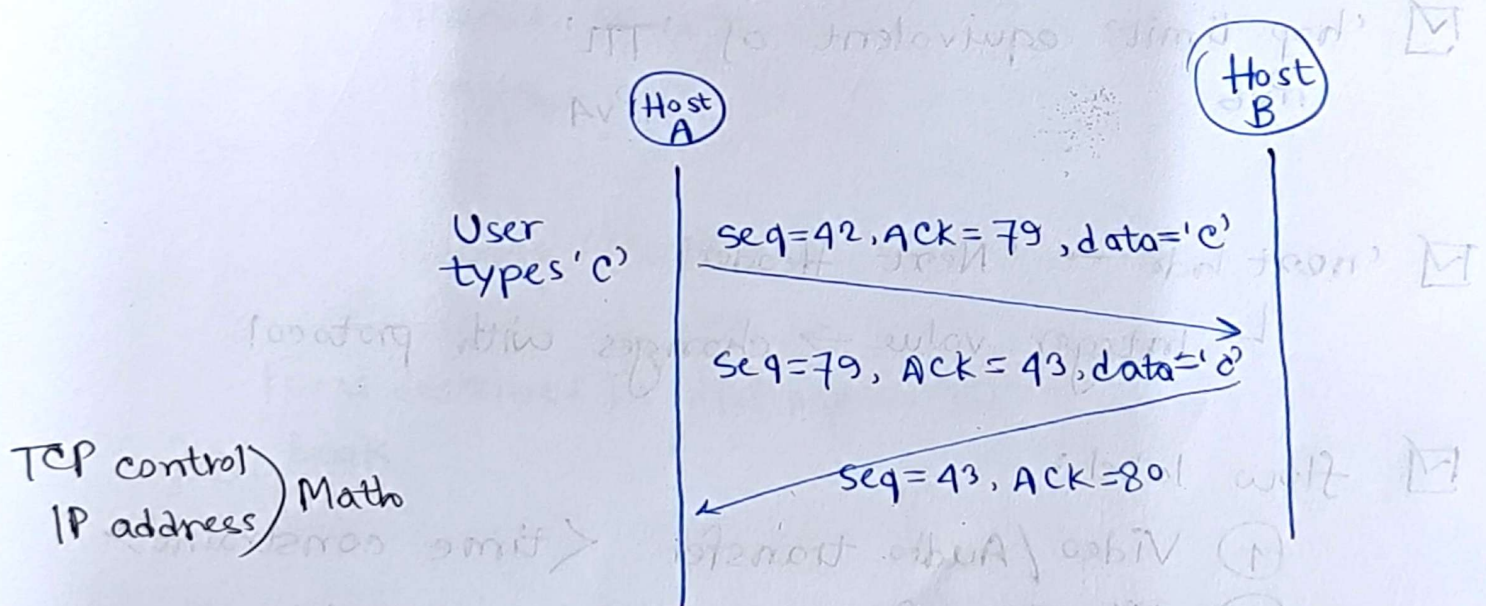
TCP/UDP	important
IPv4/IPv6	Packet Structure

{TCP, IP, HTTP, Protocols} আবেশন

TCP Segment Structure



TCP sequence Numbers, Acks



#TCP Congestion control : AIMD

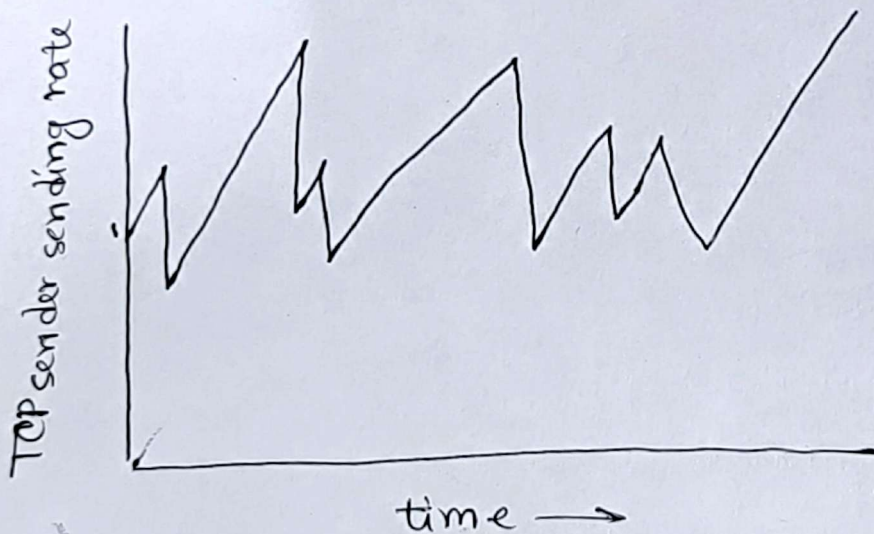
① Additive Increase (AI)

② Multiplicative Decrease (MD)

↳ cut sending rate in half at each loss event

→ increase sending rate by 1 maximum segment size every RTT until loss detected

AIMD saw tooth behavior:
probing for bandwidth



senders can increase sending rate until packet loss (congestion) occurs, then decrease sending rate on loss event.

TCP AIMD; more

Multiplicative Decrease Detail:

Sending rate is,

- ① Cut in half on loss detected by triple duplicate ACK (TCP Reno)
- ② Cut to 1 MSS (Max Segment Size) when loss detected by timeout. (TCP Tahoe)

Why AIMD?

↳ A distributed and asynchronous algorithm.

- ✓ optimizes congested flow rates network wide
- ✓ Has desirable stability properties.

Extras:

① TCP slow starts ^{phase}

↳ actually exponentially fast (2^n)

② Congestion Avoidance

③ First Recovery

$\left\{ \begin{array}{l} \text{cwnd} \rightarrow \text{congestion} \\ \text{window} \end{array} \right\}$